

$$\theta_{min} \approx \frac{\lambda}{b} = \frac{550 \times 10^{-9}}{1.8 \times 10^{-3}} = 3.05556 \times 10^{-4} rad$$

Approximating using “s = rθ”,

$$D \approx L_{max} \theta_{min}$$

$$0.77 \approx L_{max} (3.05556 \times 10^{-4})$$

$$L_{max} \approx 2519.996 = 2500 \text{ m (round DOWN, since question asks for Max. L)}$$